

Part 3

Methodology for Empirical Research





What is methodology?



The San José State University Writing Center puts it this way -
the methodology section describes how your research was conducted, so readers can check whether your approach is accurate and dependable. A good methodology increases trust in your findings.

In plain terms though ... **methodology is the heart of your paper. It's where you show exactly how you got your data, step by step.**

You are answering these four things!!

- *What evidence will be used*
- *Where that evidence comes from*
- *How it was collected*
- *How it was analyzed*
-

Empirical papers and literature reviews both need a methodology, but they look very different. **A lit review methodology is short.** *An empirical one is detailed, because someone has to be able to repeat what you did*



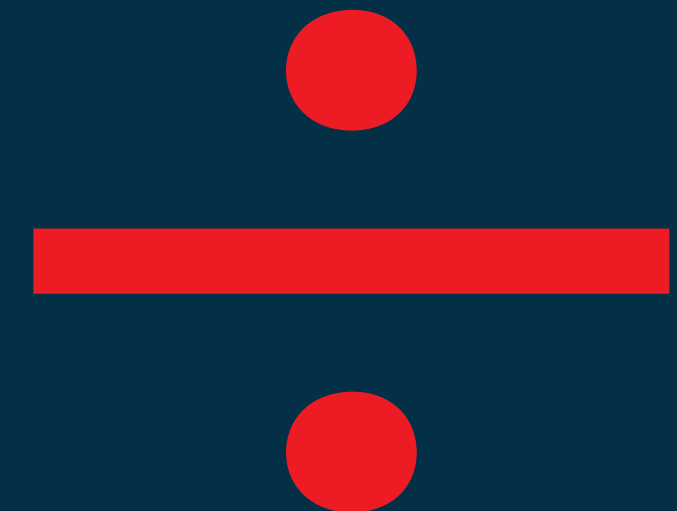
Where the paths divide?

Two students are both studying AI in education.
Student A reads 30 scholarly articles and writes a paper about what the field already knows.

Student B surveys 200 students and reports what those students actually said.

Are they using the same methodology? Never. Student A is doing a literature review. Student B is doing empirical research, meaning research based on data you collect yourself from the real world.

Different research questions demand different methods. Your method is not!!! a style choice. It follows from what you're asking.





Literature review vs empirical paper

Literature Review	Empirical Paper
Uses existing studies	Collects original data
No participants	Participants required
Sources are journal articles	Sources are survey or interview responses
Synthesizes previous findings	Produces new findings
Methodology is short	Methodology is detailed
No ethics approval usually needed	Consent and ethics matter



Why methodology matters?

Without a methodology, findings cannot be verified, the research cannot be replicated, and the conclusions are not trustworthy. **It becomes an essay you wrote for your friends, not a work for academia.**

The test to keep in mind - ***could a stranger, reading only your methodology, repeat your study and expect similar results? If no, it isn't finished.***

This is also the section reviewers attack first. A weak methodology sinks a paper faster than a weak conclusion, because if the method is flawed, the results mean nothing regardless of how interesting they sound.



The full landscape of empirical methods

Empirical research is bigger than surveys. **You almost certainly won't use most of these, but you should know they exist, because you'll read papers that used them.**

- **Experimental.** Change one thing on purpose, keep everything else constant, and measure the effect. Randomized controlled trials in medicine are the gold standard version.
- **Laboratory and wet-lab work.** Cell cultures, chemical assays, animal models, tissue samples. Requires lab access, supervision, and often ethics board approval.
- **Computational and simulation studies.** Common in computer science and physics. You build or test a model, run it on data, and report performance. Training a machine-learning model on a public dataset counts.
- **Observational studies.** Watch and record without intervening. Classroom observation, clinical cohort studies, field biology.
- **Secondary data analysis.** Take a large existing dataset (government statistics, public health records, an open Kaggle dataset) and analyze it in a new way. This one is genuinely accessible to you, and worth remembering.
- **Case studies.** Deep examination of one person, school, company, or event.
- **Content analysis.** Systematically code text, media, or images (for example, analyzing 200 news headlines for framing).



Why surveys are your best route right now?

Surveys are the most practical form of empirical research for a high-school researcher, and why?

1. **No lab, no equipment, no budget. Google Forms is free.** Right?
2. You control the timeline. A survey can go out and close in two weeks.
3. You can reach your own population easily. Your school, your peers, your community.
4. The analysis is manageable. Percentages, averages, and simple comparisons are enough for a solid paper.
5. The write-up is well established. There's a standard structure, which is exactly what we're covering today.

The tradeoff to be honest about surveys tells **you what people report about themselves, not what they actually do.** *Self-reported data can be biased by memory, honesty, and how you phrased the question. Good papers acknowledge this in their limitations!!! As we stated before*



Interviews ---- the other accessible option

If surveys give you breadth... interviews give you what? **Depth!**

A survey asks 200 people one shallow question each.

An interview asks 8 people twenty deep questions each.

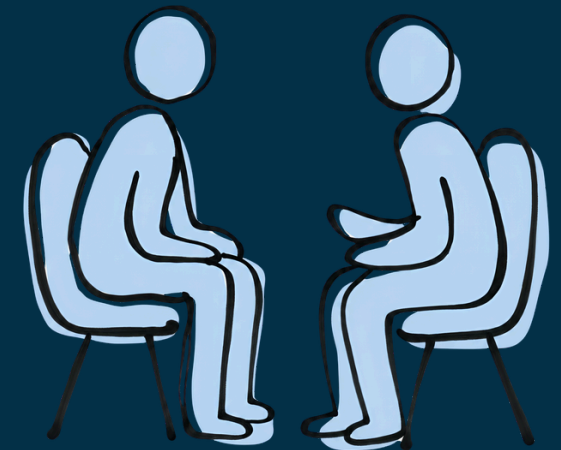
*Use interviews when you want to understand why people think something, not just how many think it. **They're the qualitative counterpart to the quantitative survey.***

Practical notes you guys might need !

interviews take much longer per participant, need recording and transcription, and are harder to analyze because you're working with paragraphs instead of numbers. But 6 to 10 good interviews is a legitimate study.

Best of both - **a mixed-methods design. Survey 150 people for the pattern, then interview 8 of them to explain the pattern.**

This is genuinely impressive at your level and not much harder!!!





The six components of a survey methodology

Every complete survey methodology section includes all six. Miss one and we will notice surely!

1. **Research design. What type of study was this?**
2. **Population. Who is the study about?**
3. **Sample. Who actually participated?**
4. **Instrument. What did you ask, and why?**
5. **Data collection. When, where, and how?**
6. **Ethical considerations. How did you protect participants?**

Component 1: Research design

The methodology always opens by naming the type of study.

Example: "This study employed a quantitative, survey-based research design to investigate how artificial intelligence influences study habits among high-school students."

Three things this sentence must make clear:

- What type of evidence was collected (numerical, textual, or both)
- Why that method was chosen over alternatives
- How it addresses your research question

Vocabulary worth knowing: quantitative means numbers and measurable patterns. Qualitative means words, meaning, and experience. Mixed-methods combines both. A Likert-scale survey is quantitative. An interview is qualitative. A survey with open-ended questions attached is mixed.

Component 2: Population

The population is the entire group you want to understand and generalize to.
Examples: high-school students in Uzbekistan. FLEX program participants in Central Asia. Teachers at international schools in Tashkent.

Important point to remember !

researchers almost never survey an entire population. The population simply defines who the study is about, and therefore who your conclusions apply to.

Example sentence: "The target population consisted of high-school students between the ages of 15 and 18."

Be honest about scope. If you only surveyed your own school, your population is not "students worldwide." Narrowing this correctly makes your paper stronger, not weaker.

Component 3: Sample

The sample is who actually completed your survey!!!!

Population: 10,000 students. Sample: 120 students.

Report all of these please

- Number of participants
- Age range
- Gender distribution, if relevant to your question
- Schools or institutions involved
- Location
- How you recruited them

Example sentence: "A convenience sample of 120 students from three secondary schools participated in the study."

Sampling terms to know ---- **a convenience sample is whoever was reachable, which is what most of you will use, and it's acceptable as long as you say so. A random sample means everyone in the population had an equal chance of selection, which is stronger but harder to achieve. Stratified sampling means deliberately including proportions of subgroups, for example equal numbers from each grade.**

Rough guidance on size: aim for at least 100 responses for a quantitative survey if you can. Below about 30, your percentages get unstable and a couple of responses can swing your entire finding.



Component 4: The instrument

This is the longest part of your methodology, and the part students at Terra most often rush

You must explain these always!!!

1. What was measured
2. How many questions the survey contained
3. What question types you used
4. Why those specific questions were chosen

Example to look at "The survey contained 18 items across three sections: demographic questions, AI usage frequency questions, and perception questions measured on a five-point Likert scale, followed by two open-ended items."

Do not just say "we made a survey." Describe its architecture. **If you adapted questions from a published study, say which one and cite it, because borrowing a validated instrument is a strength worth advertising.**



Component 5: Types of survey questions

The most interesting part, and where most papers are won or lost.

Multiple choice. Good for facts and categories.

"Which AI tool do you use most often? A) ChatGPT B) Claude C) Gemini D) I don't use AI tools"

Likert scale. Good for attitudes and opinions. Gives you numbers you can average.

"AI improves my learning." 1 = Strongly disagree, 5 = Strongly agree.

Ranking. Good for priorities.

"Rank these study methods from most to least useful."

Frequency. Good for behavior.

"How often do you use AI for homework? Daily, a few times a week, a few times a month, never."

Open-ended. Good for reasons and surprises.

"Describe one advantage of AI for learning."

A real answer you might get - **"Oh it's a great way to cheat during homework assignments." That's exactly the kind of honest response open-ended questions exist to capture, and often the most quotable material in your results.**

Balance? **mostly closed questions for analyzable data, with two or three open-ended ones for depth. All open-ended is a nightmare to analyze. All closed is bloodless.**